SURVIVAL MODELS

Data files

We will make use of three data sets. There is a DATASETS section of the module web site. There you can find information on the data sets, the files themselves and the R scripts for reading the files. You should download the files and scripts. Note that R looks for files in

```
My Documents\Rstats\Data Analysis
```

so it is a good idea to put the files here (or in a subfolder of `Data Analysis` for a particular project (see **Creating project folders** below)).

CMI data

The Continuous Mortality Investigation Bureau (CMI) collects data from UK insurance companies and investigates the mortality of the insured lives in the data. The files `CMI_Deaths.csv` and `CMI_Exposures.csv` contain death counts and central exposures for male assured lives. The R-script `CMI_read.r` reads the data files and creates two matrices of deaths and exposures, `Dth` and `Exp` respectively, for years 1950 to 2005 and ages 20 to 90. The ages and years are held in `Age` and `Year` respectively. You can check the data with `Dth[1:5, 1:5]` for example. You can select the data for particular ages and years by, for example,

```
Dth.65 = Dth[Age == 65, ]
Dth.2000 = Dth[ ,Year == 2000]
```

with corresponding code for exposure.

Human Mortality Database

The Human Mortality Database at

```
www.mortality.org\
```

is a comprehensive resource for human mortality data. We will make use of the data for England and Wales for both male and female. The files `EngWales_Deaths.txt` and `EngWales_Exposures.txt` contain death counts and central exposures for the population in England and Wales. The R-script `EngWales_read.r` reads the data files and creates four matrices, `Dth.M` and `Exp.M` of deaths and exposures for males, and `Dth.F` and `Exp.F` of deaths and exposures for females, both for ages 10 to 90 and for years 1841 to 2005. The ages and years are (also) held in `Age` and `Year` respectively.

Creating project folders

It is a good idea to keep work for different projects in separate folders. For example, suppose you want to create a subfolder `Project1` for the first project. There are two steps.

(a) Create `Project1` as a subfolder of `My Documents\Rstats\Data Analysis`.

(b) You must now make R look in this folder. Before starting R, highlight the R-icon and R-click. This bring up a menu: select *Properties*. Edit the *Start in* entry to

```
...Data Analysis\Project1
```

Now when R starts it will look in this folder for data, R-code, etc.